SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
Constituent of Symbiosis International (Deemed University), Pune
(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of india)
॥वसुधैव कुटुम्बकम्॥ Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

ASSIGNMENT No : 5

TITLE : Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Name: Arpan Gupta	Class: CSE A2
PRN: 25070122046	Batch: 2025-29

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Outcome

```java
package lab;

import java.util.ArrayList;

public class arraylist_example {
    Run | Debug
    public static void main(String args[]) {
        ArrayList<String> fruits = new ArrayList<>();

        fruits.add(e: "Apple");
        fruits.add(e: "Banana");
        fruits.add(e: "Cherry");

        System.out.println("Fruits: " + fruits);
        fruits.remove(o: "Banana");
        System.out.println("After removing Banana: " + fruits);

        System.out.println("Size: " + fruits.size());
        System.out.println("Contains Apple? " + fruits.contains(o: "Apple"));
    }
}
```

```java
package lab;

import java.util.Vector;

public class vector_Example {
    Run | Debug
    public static void main(String args[]) {
        Vector<Integer> numbers = new Vector<>();

        numbers.add(e: 100);
        numbers.add(e: 200);
        numbers.add(e: 300);

        System.out.println("Vector: " + numbers);
        numbers.remove(index: 1);
        System.out.println("After removing index 1: " + numbers);

        System.out.println("Size: " + numbers.size());
        System.out.println("Element at index 0: " + numbers.get(index: 0));
    }
}
```

```java
package lab;

public class stringbuffer_example {
    Run | Debug
    public static void main(String args[]) {
        StringBuffer sb = new StringBuffer();

        sb.append(str: "Hello");
        sb.append(str: " ");
        sb.append(str: "World");

        System.out.println("StringBuffer: " + sb);
        sb.delete(start: 5, end: 6);
        System.out.println("After deleting space: " + sb);

        System.out.println("Length: " + sb.length());
        System.out.println("Character at index 0: " + sb.charAt(index: 0));
    }
}
```

Conclusion

This assignment demonstrates the concepts of ArrayList, Vector and StringBuffer, as well as demonstrates the use of these classes in applications requiring dynamic memory allocation and efficient string processing.

Exercise

1. Create a **To-Do List application** using ArrayList to store tasks and StringBuffer to display tasks.

todolist_example.java

Something > src > lab > todolist_example.java > todolist_example

```java
package lab;

import java.util.ArrayList;
import java.util.Scanner;

public class todolist_example {
    Run | Debug
    public static void main(String args[]) {
        ArrayList<String> tasks = new ArrayList<>();
        Scanner scanner = new Scanner(System.in);
        int choice;

        System.out.println(x: "===== TO-DO LIST APPLICATION =====");

        do {
            // Display menu using StringBuffer
            StringBuffer menu = new StringBuffer();
            menu.append(str: "\n--- Menu ---\n");
            menu.append(str: "1. Add Task\n");
            menu.append(str: "2. Remove Task\n");
            menu.append(str: "3. List All Tasks\n");
            menu.append(str: "4. Exit\n");
            menu.append(str: "Enter your choice: ");
            System.out.print(menu);

            choice = scanner.nextInt();
            scanner.nextLine(); // consume newline

            switch (choice) {
                case 1:
                    // Add task
                    System.out.print(s: "Enter task to add: ");
                    String newTask = scanner.nextLine();
                    tasks.add(newTask);
                    System.out.println(x: "Task added successfully!");
                    break;

```

```
                case 2:
                    // Remove task
                    if (tasks.isEmpty()) {
                        System.out.println(x: "No tasks to remove.");
                    } else {
                        // Display tasks with index
                        StringBuffer taskList = new StringBuffer();
                        taskList.append(str: "Current Tasks:\n");
                        for (int i = 0; i < tasks.size(); i++) {
                            taskList.append(i + 1).append(str: ". ").append(tasks.get(i)).append(str: "\n");
                        }
                        System.out.print(taskList);
                        System.out.print(s: "Enter task number to remove: ");
                        int taskNum = scanner.nextInt();
                        scanner.nextLine(); // consume newline

                        if (taskNum >= 1 && taskNum <= tasks.size()) {
                            String removedTask = tasks.remove(taskNum - 1);
                            System.out.println("Removed: " + removedTask);
                        } else {
                            System.out.println(x: "Invalid task number!");
                        }
                    }
                    break;

                case 3:
                    // List all tasks using StringBuffer
                    if (tasks.isEmpty()) {
                        System.out.println(x: "No tasks in the list.");
                    } else {
                        StringBuffer displayTasks = new StringBuffer();
                        displayTasks.append(str: "\n===== YOUR TASKS =====\n");
                        for (int i = 0; i < tasks.size(); i++) {
                            displayTasks.append(i + 1).append(str: ". ");
                            displayTasks.append(tasks.get(i));
                            displayTasks.append(str: "\n");
                        }
                        displayTasks.append(str: "====================\n");
                        displayTasks.append(str: "Total tasks: ").append(tasks.size()).append(str: "\n");
                        System.out.print(displayTasks);
                    }
                    break;
```

```
                case 4:
                    System.out.println(x: "Exiting To-Do List Application. Goodbye!");
                    break;

                default:
                    System.out.println(x: "Invalid choice! Please enter a number between 1 and 4.");
            }
        } while (choice != 4);

        scanner.close();
    }
}
```

2. Create a **Student Course Registration System** using **ArrayList** to store the list of courses registered by a student and **StringBuffer** to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

CourseRegistration.java

Something > src > lab > CourseRegistration.java > CourseRegistration

```java
package lab;

import java.util.ArrayList;
import java.util.Scanner;

public class CourseRegistration {
    Run | Debug
    public static void main(String args[]) {
        ArrayList<String> courses = new ArrayList<>();
        Scanner scanner = new Scanner(System.in);
        int choice;

        System.out.println(x: "===== STUDENT COURSE REGISTRATION SYSTEM =====");

        do {
            // Display menu using StringBuffer
            StringBuffer menu = new StringBuffer();
            menu.append(str: "\n--- Menu ---\n");
            menu.append(str: "1. Add Course\n");
            menu.append(str: "2. Remove Course\n");
            menu.append(str: "3. View Registered Courses\n");
            menu.append(str: "4. Exit\n");
            menu.append(str: "Enter your choice: ");
            System.out.print(menu);

            choice = scanner.nextInt();
            scanner.nextLine(); // consume newline

            switch (choice) {
                case 1:
                    // Add course
                    System.out.print(s: "Enter course name to register: ");
                    String newCourse = scanner.nextLine();
                    courses.add(newCourse);
                    System.out.println("Course \"" + newCourse + "\" registered successfully!");
                    break;

```

```
37                  case 2:
38                      // Remove course
39                      if (courses.isEmpty()) {
40                          System.out.println(x: "No courses registered to remove.");
41                      } else {
42                          // Display courses with index using StringBuffer
43                          StringBuffer courseList = new StringBuffer();
44                          courseList.append(str: "Registered Courses:\n");
45                          for (int i = 0; i < courses.size(); i++) {
46                              courseList.append(i + 1).append(str: ". ").append(courses.get(i)).append(str: "\n");
47                          }
48                          System.out.print(courseList);
49                          System.out.print(s: "Enter course number to remove: ");
50                          int courseNum = scanner.nextInt();
51                          scanner.nextLine(); // consume newline
52
53                          if (courseNum >= 1 && courseNum <= courses.size()) {
54                              String removedCourse = courses.remove(courseNum - 1);
55                              System.out.println("Removed course: " + removedCourse);
56                          } else {
57                              System.out.println(x: "Invalid course number!");
58                          }
59                      }
60                      break;
61
```

```
62                  case 3:
63                      // View all registered courses using StringBuffer
64                      if (courses.isEmpty()) {
65                          System.out.println(x: "No courses registered yet.");
66                      } else {
67                          StringBuffer displayCourses = new StringBuffer();
68                          displayCourses.append(str: "\n==================================\n");
69                          displayCourses.append(str: "     REGISTERED COURSES\n");
70                          displayCourses.append(str: "==================================\n");
71                          for (int i = 0; i < courses.size(); i++) {
72                              displayCourses.append(i + 1).append(str: ". ");
73                              displayCourses.append(courses.get(i));
74                              displayCourses.append(str: "\n");
75                          }
76                          displayCourses.append(str: "==================================\n");
77                          displayCourses.append(str: "Total courses registered: ").append(courses.size()).append(str: "\n");
78                          displayCourses.append(str: "==================================\n");
79                          System.out.print(displayCourses);
80                      }
81                      break;
82
83                  case 4:
84                      System.out.println(x: "Exiting Student Course Registration System. Goodbye!");
85                      break;
86
87                  default:
88                      System.out.println(x: "Invalid choice! Please enter a number between 1 and 4.");
89              }
90          } while (choice != 4);
91
92          scanner.close();
93      }
94  }
```